

Visco-Elastic Cycle Models

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1 Equations

We will use tensor deviatoric plus volumetric decomposition for the stress tensor

$$\boldsymbol{\sigma} = \boldsymbol{s} + p\boldsymbol{I},$$

and the strain tensor

$$\boldsymbol{\epsilon} = \boldsymbol{e} + \frac{1}{3}\epsilon_v\boldsymbol{I},$$

where

$$\epsilon_v(\boldsymbol{u}) = \nabla \cdot \boldsymbol{u}, \quad \boldsymbol{e}(\boldsymbol{u}) = \frac{1}{2}(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T) - \frac{1}{3}\epsilon_v(\boldsymbol{u})\boldsymbol{I}.$$

We use the following definitions for the deviatoric stress $\boldsymbol{s}$ and the pressure (volumetric stress) via

$$\boldsymbol{s} = 2G\boldsymbol{e}, \quad p = K\epsilon_v,$$

where G is the shear modulus and K is the bulk modulus.

Coefficient helpers:

$$K = \frac{E}{3(1-2\mu)}, \quad G = \mu = \frac{E}{2(1+\mu)} \quad \text{should be Poisson's ratio}$$

and

$$\mu = G, \quad \lambda = K - \frac{2}{3}G$$

and the Poisson's ratio is

$$\nu = \frac{3K - 2G}{6K + 2G} = \frac{\lambda}{2(\lambda + \mu)}.$$

The conservation of momentum ($\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$) becomes

$$2G\nabla \cdot \mathbf{e}(\mathbf{u}) + K\nabla\epsilon_v(\mathbf{u}) = \mathbf{0}.$$

In the visco-elastic model we assume K is constant, e.g. viscous effects are confined to only the shear stress components, thus we have

$$\frac{1}{2G}\dot{\mathbf{s}} + \frac{1}{2\eta}\mathbf{s} = \dot{\mathbf{e}} = \mathbf{F}.$$

Discretize the above constitutive equation with implicit Euler, leading to

$$\frac{\mathbf{s}^{k+1} - \mathbf{s}^k}{\Delta t} + \frac{G}{\eta}\mathbf{s}^{k+1} = 2G\frac{\mathbf{e}^{k+1} - \mathbf{e}^k}{\Delta t}.$$

Doing some algebra (steps included to ensure I didn't screw it up) we have

$$\mathbf{s}^{k+1} + \Delta t \frac{G}{\eta} \mathbf{s}^{k+1} = 2G(\mathbf{e}^{k+1} - \mathbf{e}^k) + \mathbf{s}^k$$

and

$$1 + \Delta t \frac{G}{\eta} = \frac{\eta + \Delta t G}{\eta}$$

and

$$\mathbf{s}^{k+1} + \Delta t \frac{G}{\eta} \mathbf{s}^{k+1} = 2G\mathbf{e}^{k+1} - 2G\mathbf{e}^k + \mathbf{s}^k.$$

Finally we have

$$\mathbf{s}^{k+1} = \left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^{k+1} - 2G\mathbf{e}^k + \mathbf{s}^k).$$

Now the conservation of momentum $\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$ becomes

$$\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}^{k+1}) = \nabla \cdot \mathbf{s}(\mathbf{u}^{k+1}) + K\nabla\epsilon_v(\mathbf{u}^{k+1}) \tag{1}$$

$$= \nabla \cdot \left[\left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^{k+1} - 2G\mathbf{e}^k + \mathbf{s}^k) \right] + K\nabla\epsilon_v(\mathbf{u}^{k+1}) \tag{2}$$

$$= \mathbf{0} \tag{3}$$

thus we have

$$\nabla \cdot \left[\left(\frac{2G\eta}{\eta + \Delta t G} \right) \mathbf{e}^{k+1} \right] + K\nabla\epsilon_v(\mathbf{u}^{k+1}) = \nabla \cdot \left[\left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^k - \mathbf{s}^k) \right],$$

which we may write as

$$\nabla \cdot [2G'(\Delta t)\mathbf{e}^{k+1}] + K\nabla\epsilon_v(\mathbf{u}^{k+1}) = \nabla \cdot \left[\left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^k - \mathbf{s}^k) \right],$$

which looks like standard elasticity formulation with a modified shear modulus $G'(\Delta t)$ and a non-zero right side, where

$$G'(\Delta t) = G\alpha(\Delta t), \quad \alpha(\Delta t) = \frac{\eta}{\eta + \Delta t G}.$$

Stress Update

•

$$\mathbf{s}_{ve}^{k+1} = \text{dev}(\mathcal{F}(G\alpha(\Delta t), K))$$

•

$$\mathbf{s}^{k+1} = \mathbf{s}_{ve}^{k+1} - \text{dev}(\mathcal{F}(G\alpha(\Delta t), K)) + \alpha(\Delta t)\mathbf{s}^k$$

where $\mathcal{F}(\cdot)$ is the Green's function.

Better explanation

Let S^k denote the slip at time t^k and let

$$\boldsymbol{\sigma}^k = \mathcal{F}(G, K, S^k)$$

define the Green's function which maps slip S^k to stress $\boldsymbol{\sigma}^k$, that is $\mathcal{F}$ provides an analytic solution to

$$\nabla \cdot \boldsymbol{\sigma}^k = \mathbf{0}, \quad \text{jump}(\mathbf{u}) = S^k \quad \text{on } \Gamma_{\text{fault}}$$

where $\boldsymbol{\sigma}^k = 2G\mathbf{e}^k + K\epsilon_v^k\mathbf{I}$. Hence to compute the stress update on the fault Γ_{fault}

$$\boldsymbol{\sigma}^{k+1} = 2G'(\Delta t)\mathbf{e}^{k+1} + K\epsilon_v^{k+1}\mathbf{I} - 2G'(\Delta t)\mathbf{e}^k + \alpha(\Delta t)\mathbf{s}^k$$

we do

$$\boldsymbol{\sigma}^{k+1} = \mathcal{F}(G'(\Delta t), K, S^{k+1}) - \text{dev}(\mathcal{F}(G'(\Delta t), K, S^k)) + \alpha(\Delta t)\mathbf{s}^k. \quad \text{eq I}$$

This update requires storing the slip S^k from the previous time step along with $\mathbf{s}^k$ (the deviatoric stress). One might be tempted to store the result

of $\text{dev}(\mathcal{F}(G'(\Delta t), K, S^k))$, however this would be unwise as Δt will change from step k to $k + 1$.

It would be more explicit to define a step k dependent time step, i.e. Δt^k . Then $t^{k+1} = t^k + \Delta t^{k+1}$ so we would have

$$\boldsymbol{\sigma}^{k+1} = \mathcal{F}(G'(\Delta t^{k+1}), K, S^{k+1}) - \text{dev}(\mathcal{F}(G'(\Delta t^{k+1}), K, S^k)) + \alpha(\Delta t^{k+1})\mathbf{s}^k.$$

Note that in the very short time step limit $\Delta t^{k+1} \rightarrow 0$ the elastic response is recovered since

$$\alpha(\Delta t^{k+1}) \rightarrow 1, \quad G' \rightarrow G$$

and thus

$$-\text{dev}(\mathcal{F}(G'(\Delta t^{k+1}), K, S^k)) + \alpha(\Delta t^{k+1})\mathbf{s}^k \rightarrow -\text{dev}(\mathcal{F}(G, K, S^k)) + \mathbf{s}^k = 0$$

leaving behind

$$\boldsymbol{\sigma}^{k+1} = \mathcal{F}(G, K, S^{k+1}).$$

2 Alternative VE models

Isotropic 1D visco-elastic material would have a stress $\sigma(t)$ given by

$$\sigma(t) = \int_0^t E(t - \hat{\tau}) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau},$$

where ϵ is the strain and where the relaxation modulus $E(t)$ may be expanded as a Phony series given by

$$E(t) = E_\infty + \sum_{i=1}^N E_i \exp(-t/\tau_i).$$

In the above $\tau_i = \eta_i/E_i$ is the relaxation time of the i -th Maxwell element. Then sub-ing in we have

$$\sigma(t) = E_\infty \epsilon(t) + \sum_{i=1}^N h_i(t),$$

where $h_i(t)$ are the internal state variables (history variables) for stress. Specifically we have

$$h_i(t) = \int_0^t E_i \exp(-(t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau}$$

To find the state at $t^{k+1} = t^k + \Delta t$ we evaluate the history integral $h_i(t^{k+1})$ by splitting the time domain:

$$h_i(t^{k+1}) = \int_0^{t^k} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} + \int_{t^k}^{t^{k+1}} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau}$$

which can be re-written as

$$h_i(t^{k+1}) = \exp(-\Delta t/\tau_i) \int_0^{t^k} E_i \exp(-(t^k - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} + \int_{t^k}^{t^{k+1}} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} \quad (4)$$

$$= \exp(-\Delta t/\tau_i) h_i(t^k) + \int_{t^k}^{t^{k+1}} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} \quad (5)$$

so in general we may write the update as

$$h_i(t^{k+1}) = \exp(-\Delta t/\tau_i) h_i(t^k) + \Delta h_i.$$

Making the assumption that the strain rate is constant over each time step Δt , i.e.

$$\frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} \approx \frac{\epsilon^{k+1} - \epsilon^k}{\Delta t} = \frac{\Delta \epsilon}{\Delta t}$$

we have

$$\Delta h_i \approx E_i \frac{\Delta \epsilon}{\Delta t} \int_{t^k}^{t^{k+1}} \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) d\hat{\tau} = E_i \tau_i (1 - \exp(-\Delta t/\tau_i)) \frac{\Delta \epsilon}{\Delta t}$$

leading to

$$h_i(t^{k+1}) = \exp(-\Delta t/\tau_i) h_i(t^k) + E_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)) \Delta \epsilon$$

Then the discrete stress update is

$$\sigma^{k+1} = E_\infty \epsilon^{k+1} + \sum_{i=1}^N h_i(t^{k+1}) \quad (6)$$

$$= \sigma^k + \tilde{E} \Delta \epsilon + \sum_{i=1}^N (\exp(-\Delta t/\tau_i) - 1) h_i(t^k) \quad (7)$$

where

$$\tilde{E} = E_\infty + \sum_{i=1}^N E_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)).$$

Moving to 3D

$$G(t) = G_\infty + \sum_{i=1}^N G_i \exp(-t/\tau_i), \quad K(t) = K_\infty$$

$$\mathbf{h}_i(t^{k+1}) = \exp(-\Delta t/\tau_i) \mathbf{h}_i(t^k) + 2G_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)) \Delta \mathbf{e}$$

where

$$\mathbf{e} = \epsilon - \frac{1}{3} \nabla \cdot \mathbf{u} \mathbf{I}, \quad \Delta \mathbf{e} = \mathbf{e}^{k+1} - \mathbf{e}^k$$

As before we can define

$$\tilde{G} = G_\infty + \sum_{i=1}^N G_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)).$$

and letting $\boldsymbol{\sigma} = \tilde{\mathbb{C}} \boldsymbol{\epsilon}$ we have

$$\tilde{\mathbb{C}} = K_\infty (\mathbf{I} \otimes \mathbf{I}) + 2\tilde{G} \left(\mathbb{I} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I} \right)$$

The stress update finally becomes

$$p^{k+1} = K_\infty \text{Tr}(\epsilon^{k+1})$$

and

$$\mathbf{s}^{k+1} = 2G_\infty \mathbf{e}^{k+1} + \sum_{i=1}^N \mathbf{h}_i^{k+1}$$

and

$$\boldsymbol{\sigma}^{k+1} = \mathbf{s}^{k+1} + p^{k+1} \mathbf{I}.$$

3 For Kaj - Where does the hereditary integral go?

Elastic

$$\tau_i(t) = \tau_{0,i} + \sum_{j=1}^N K_{ij}(t) [\delta_j(t) - V_{pl,j} t] - \eta V_i(t)$$

where K_{ij} is the dense GF matrix, i is the slip patch index.

For a visco-elastic case, the interaction kernel K_{ij} is replaced by a time-dependent kernel $K_{ij}(t - t')$, resulting in the hereditary integral

$$\tau_i(t) = \tau_{0,i} + \sum_{j=1}^N \int_0^t K_{ij}(t - t') \frac{\partial \delta_j(t')}{\partial t} dt' - \eta V_i(t) \quad (8)$$

where $\frac{\partial \delta_j(t')}{\partial t}(t') = V_j(t')$ is the slip velocity history of the patch. We note the $V_{pl,j}t$ will reappear in the background stress.

One can find

$$K_{ij}(t-t') = K_{ij}^{\text{elastic}} \delta_{\text{dirac}}(t-t') + K_{ij}^{\text{viscous}} \exp\left(-\frac{t-t'}{\tau_m}\right), \quad \tau_m = \eta_{\text{visc}}/G. \quad (9)$$

Evaluating the full hereditary integral at every time step requires saving the slip velocity history of all patches for all past time steps. This scales as $\mathcal{O}(N \times M)$ where M is the number of time steps. Fixing a temporal window of length $m' < M$ steps obviously reduces (and limits) the required memory.

Since the Maxwell relaxation follows an exponential decay $\exp(-t/\tau_m)$, you can avoid storing the entire history by introducing a viscous stress state variable, $H_{ij}(t)$, for every patch interaction

$$\tau_i(t) = \tau_{0,i} + \sum_{j=1}^N K_{ij}^{\text{elastic}} [\delta_j(t) - V_{pl,j}t] + \sum_{j=1}^N H_{ij}(t) - \eta V_i(t). \quad (10)$$

Instead of integrating over past history, you update H_{ij} dynamically at each time step using its differential form

$$\frac{dH_{ij}}{dt} = -\frac{H_{ij}}{\tau_m} + K_{ij}^{\text{viscous}} V_j(t). \quad (11)$$

If you didn't want to solve this additional ODE for the stress history, you could make an approximation by assuming that $V_j(t)$ is constant over a time-step Δt and then just directly integrate the LHS and RHS w.r.t. time t yielding

$$H_{ij}(t + \Delta t) = H_{ij}(t) \exp\left(-\frac{\Delta t}{\tau_m}\right) + K_{ij}^{\text{viscous}} \Delta \delta_j \left(\frac{1 - \exp(-\Delta t/\tau_m)}{\Delta t/\tau_m}\right),$$

where $\Delta\delta_j = \hat{V}_j\Delta t$ and $\hat{V}_j$ is the assumed constant value of $V_j(t)$ over time $[t, t + \Delta t]$.

To see where this comes from, consider

$$\frac{dH}{dt} = -\frac{H}{\tau} + KV, \quad KV = \text{const.}$$

$$H(t) = KV\tau + (H_0 - KV\tau)e^{-t/\tau} = H_0e^{-t/\tau} + KV\tau(1 - e^{-t/\tau})$$

$$H(t) = H_0e^{-t/\tau} + K\frac{S}{\Delta t}\tau(1 - e^{-t/\tau})$$

To move from $t = 0$ where H_0 is defined, we evaluate the above equation at $t = \Delta t^{(1)} = t^{(1)}$ to give H_1 we have

$$H(\Delta t^{(1)}) = H_1 = H_0e^{-\Delta t^{(1)}/\tau} + K\frac{S}{\Delta t^{(1)}}\tau(1 - e^{-\Delta t^{(1)}/\tau})$$

The next step $t^{(2)} = t^{(1)} + \Delta t^{(2)}$ is given by

$$H(t^{(2)}) = H(t^{(1)} + \Delta t^{(2)}) = H_2 = H_1e^{-\Delta t^{(2)}/\tau} + K\frac{S}{\Delta t^{(2)}}\tau(1 - e^{-\Delta t^{(2)}/\tau}).$$

Note that each time-step k can be thought as a single integration moving from a reference state k at time $t^{(k)}$ to a new state which is advanced by $\Delta t^{(k)}$. For example at step 1 we move from $t = 0$ (reference state) to $t = \Delta t^{(1)}$ (new state) and for step 2 we move from state $t = t^{(1)}$ (reference state) to the state at $t = t^{(2)} = t^{(1)} + \Delta t^{(2)}$.

Derivation Insert Eq. (9) into Eq. (8) and evaluate the integrals. The first integral is

$$\sum_{j=1}^N \int_0^t K_{ij}^{\text{elastic}} \delta_{\text{dirac}}(t - t') \frac{\partial \delta_j(t')}{\partial t} dt' = \sum_{j=1}^N K_{ij}^{\text{elastic}} \int_0^t \delta_{\text{dirac}}(t - t') V_j(t') dt' \quad (12)$$

$$= \sum_{j=1}^N K_{ij}^{\text{elastic}} V_j(t) \quad (13)$$

The above equation tells us the rate of elastic stress change occurring at time t due to the current velocity field. If we want to know the total accumulated

elastic stress change from the beginning of the simulation ($t = 0$) up to the current time (t), we integrate that velocity over the entire time history.

$$\int_0^t \left[\sum_{j=1}^N K_{ij}^{\text{elastic}} V_j(t') \right] dt' = \int_0^t \left[\sum_{j=1}^N K_{ij}^{\text{elastic}} \frac{\partial \delta_j(t')}{\partial t'} \right] dt' \quad (14)$$

$$= \sum_{j=1}^N K_{ij}^{\text{elastic}} \int_0^t \left[\frac{\partial \delta_j(t')}{\partial t'} dt' \right] \quad (15)$$

$$= \sum_{j=1}^N K_{ij}^{\text{elastic}} \delta_j(t) \quad (16)$$

Let

$$\begin{aligned} H_{ij}(t) &= \int_0^t K_{ij}^{\text{viscous}} \exp\left(-\frac{t-t'}{\tau_m}\right) V_j(t') dt' \\ &= \exp\left(-\frac{t}{\tau_m}\right) \int_0^t K_{ij}^{\text{viscous}} \exp\left(\frac{t'}{\tau_m}\right) V_j(t') dt' \end{aligned} \quad (17)$$

Now compute the time derivative of $H_{ij}(t)$

$$\frac{dH}{dt} = \frac{d}{dt} \left[\exp\left(-\frac{t}{\tau_m}\right) \int_0^t K_{ij}^{\text{viscous}} \exp\left(\frac{t'}{\tau_m}\right) V_j(t') dt' \right] \quad (18)$$

$$= \frac{d}{dt} [A(t)B(t)] \quad (19)$$

$$= -\frac{1}{\tau_m} \exp(-t/\tau_m) B(t) + A(t) \frac{d}{dt} \left[\int_0^t K_{ij}^{\text{viscous}} \exp\left(\frac{t'}{\tau_m}\right) V_j(t') dt' \right] \quad (20)$$

$$= -\frac{1}{\tau_m} \exp(-t/\tau_m) B(t) + A(t) K_{ij}^{\text{viscous}} \exp\left(\frac{t}{\tau_m}\right) V_j(t) \quad (21)$$

$$\frac{dH}{dt} = -\frac{1}{\tau_m} \left[\exp(-t/\tau_m) \int_0^t K_{ij}^{\text{viscous}} \exp\left(\frac{t'}{\tau_m}\right) V_j(t') dt' \right] + K_{ij}^{\text{viscous}} V_j(t) \quad (22)$$

$$= -\frac{1}{\tau_m} [H_{ij}(t)] + K_{ij}^{\text{viscous}} V_j(t) \quad (23)$$

The essential idea is that if you have an ODE, say

$$\frac{dy}{dt} = F(t), \quad y(0) = y_0$$

you can obtain the value $y(T)$ via:

1. solving the ODE analytically and evaluating the solution $y(t)$ at $t = T$, or by;
2. computing the integral

$$y(T) = y_0 + \int_0^T F(t') dt'.$$

This is the reason why methods which are used to solve ODEs are often called “integrators”.

4 The Laplace Transform way

Basic idea

Use the correspondence principal: This is done in steps (1) take the constitutive equation (2) takes its Laplace transform (3) manipulate the expression so that it's form matches the form used in pure elasticity, this will entail defining Laplace domain dependent parameters e.g. $\bar{G}(s)$ (5) compute all Lamé parameters required by Green's function using $\bar{G}(s), K$, (6) insert $\bar{\mu}(s), \bar{\lambda}(s)$ in the Green's function, (7) transform back into the time domain.

Form 1 Start with the constitutive equation for an elastic material

$$\sigma_{ij} = 2G(\epsilon_{ij} - \frac{1}{3}\epsilon_{kk}\delta_{ij}) + K\epsilon_{kk}\delta_{ij} \quad (24)$$

$$= 2G\epsilon_{ij} + (K - \frac{2G}{3})\epsilon_{kk}\delta_{ij} \quad (25)$$

$$= 2G\epsilon_{ij} + \lambda\epsilon_{kk}\delta_{ij} \quad (26)$$

where $\lambda = K - 2G/3$.

Start with the constitutive equation for a visco-elastic material

$$\frac{d\sigma_{ij}}{dt} + \frac{1}{\tau_m}(\sigma_{ij} - \frac{1}{3}\sigma_{kk}\delta_{ij}) = 2G\frac{d\epsilon_{ij}}{dt} + \lambda\frac{d\epsilon_{kk}}{dt}\delta_{ij}.$$

Denote the Laplace transform of a function $f(t)$ as

$$\mathcal{L}\{f(t)\} = \bar{f}(s).$$

Basic function transformations:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\bar{f}(s) - f(0), \quad \mathcal{L}\{c\} = \frac{c}{s}.$$

Take Laplace transform of the constitutive equation above to give

$$s\bar{\sigma}_{ij}(s) + \frac{1}{\tau_m}(\bar{\sigma}_{ij}(s) - \frac{1}{3}\bar{\sigma}_{kk}(s)\delta_{ij}) = 2Gs\bar{\epsilon}_{ij}(s) + \lambda s\bar{\epsilon}_{kk}(s)\delta_{ij}.$$

Do some algebra

$$(s + \frac{1}{\tau_m})\bar{\sigma}_{ij}(s) = 2Gs\bar{\epsilon}_{ij}(s) + \lambda s\bar{\epsilon}_{kk}(s)\delta_{ij} + \frac{1}{\tau_m}\frac{1}{3}\bar{\sigma}_{kk}(s)\delta_{ij}$$

$$\bar{\sigma}_{ij}(s) = 2G\frac{s}{s + \frac{1}{\tau_m}}\bar{\epsilon}_{ij}(s) + \frac{1}{s + \frac{1}{\tau_m}}(\lambda s)\bar{\epsilon}_{kk}(s)\delta_{ij} + \frac{1}{s + \frac{1}{\tau_m}}\left(\frac{1}{\tau_m}\frac{1}{3}\right)\bar{\sigma}_{kk}(s)\delta_{ij},$$

and then define

$$\bar{G}(s) = \frac{Gs}{s + \frac{1}{\tau_m}}.$$

We also have $K(s) = K$ because the material is not compressible in bulk.

Note to self. The Laplace transform of Form 1 introduces a term σ_{kk} which is not naturally present in the form given by Form 1. This term ONLY affects the volumetric stress. Hence Form 1 is going to be fine if you only intend to use the GF for the deviatoric stress (i.e. as required for the shear stress on the fault).

The Green's functions we work with are usually going to be written in terms of λ and ν , so we will need their form in the Laplace domain, e.g.

$$\bar{\lambda}(s) = K - \frac{2}{3}\bar{G}(s),$$

$$\bar{\nu}(s) = \frac{\bar{\lambda}(s)}{2(\bar{\lambda}(s) + \bar{\mu}(s))} = \frac{3K - 2\bar{\mu}(s)}{2(3K + \bar{\mu}(s))}$$

Write GF in the following form decomposed form which isolates spatial terms and material properties associated with either the deviatoric stress and volumetric stress

$$\sigma(G, \nu, \delta(t)) = F_1(x)G\delta(t) + F_2(x)F_2^m(G, \nu)\delta(t)$$

in which $F_1(x)G$ describes the deviatoric part of the stress and $F_2(x)F_2^m(G, \nu)$ describes the volumetric part of the stress. Folks usually employ this definition

$$F_1(x)G = K^{\text{elastic}}(x),$$

that $\tau(t) = K^{\text{elastic}}(x)\delta(t)$. Transform to the Laplace domain

$$\bar{\sigma}(\bar{G}(s), \bar{\nu}(s), \bar{\delta}(s)) = F_1(x)\bar{G}(s)\bar{\delta}(s) + F_2(x)\bar{F}_2^m(\bar{G}(s), \bar{\nu}(s))\bar{\delta}(s)$$

Time domain

$$\tau(t) = K^{\text{elastic}}\Delta\delta(t)$$

Laplace domain

$$\bar{\tau}(s) = \frac{\bar{G}(s)}{G} K^{\text{elastic}} \bar{V}(s)$$

Using $s/(s+a) = 1 - a/(s+a)$ we have

$$\bar{\tau}(s) = \left(1 - \frac{1/\tau_m}{s + 1/\tau_m}\right) K^{\text{elastic}} \bar{V}(s).$$

Collectively this leads to the following Laplace transform for the complete constitutive equation

$$\bar{\tau}(s) = K^{\text{elastic}} \bar{V}(s) - \left(\frac{1/\tau_m}{s + 1/\tau_m}\right) K^{\text{elastic}} \bar{V}(s).$$

Inverse Laplace Transform

$$\mathcal{L}^{-1}\{K^{\text{elastic}} \bar{V}(s)\} = K^{\text{elastic}} V(t)$$

Notes: Inverse transform of $1/(s+a)$ is e^{-at} , so inverse transform of $a/(s+a)$ is ae^{-at} . Multiplication of two functions of s is Laplace domain, i.e. $\bar{f}(s)\bar{g}(s)$ maps to convolution integral in the time domain. Hence we have

$$\mathcal{L}^{-1}\left\{\left(\frac{1/\tau_m}{s + 1/\tau_m}\right) \bar{V}(s)\right\} = \int_0^t \frac{1}{\tau_m} \exp\left(\frac{-(t-t')}{\tau_m}\right) V(t') dt'.$$

Finally, for the complete constitutive equation we have

$$\tau(t) = K^{\text{elastic}} V(t) - \int_0^t \left(\frac{1}{\tau_m} K^{\text{elastic}}\right) \exp\left(\frac{-(t-t')}{\tau_m}\right) V(t') dt'.$$

Folks like to use this definition:

$$K^{\text{viscous}} = -\frac{1}{\tau_m} K^{\text{elastic}}.$$

Wrapping this all up for an actual simulation, we can use Eq. (10) along with the ODE given by Eq. (11) where the definition of H is given by Eq. (17).

Form 2 Elastic:

$$\tau_{ij} = 2Ge_{ij}, \quad \frac{1}{3}\sigma_v = \frac{1}{3}K\epsilon_v \quad \longrightarrow \quad \sigma_{ij} = \tau_{ij} + \sigma_v\delta_{ij}$$

Visco-elastic with dev-vol parts

$$\frac{d\tau_{ij}}{dt} + \frac{1}{\tau_m}\tau_{ij} = 2G\frac{de_{ij}}{dt}$$

$$\frac{d\sigma_v}{dt} = K\frac{\epsilon_v}{dt}$$

Laplace transform

$$s\bar{\tau}_{ij}(s) + \frac{1}{\tau_m}\bar{\tau}_{ij} = 2Gs\bar{e}_{ij}(s)$$

$$\bar{\tau}_{ij}(s) = 2G\frac{s}{s + 1/\tau_m}\bar{e}_{ij}(s)$$

$$s\bar{\sigma}_v(s) = Ks\bar{\epsilon}_v(s)$$